

Model Lab

Every prior, same slate — 2026-09-22

SERVED PRIOR

Core (KNN+SVR)

PRIORS COMPARED

10

HELD-OUT PA

36,363

Held-Out Scores

Fitted on plate appearances before 2026-08-16, scored on the 36,363 that came after (actual HR rate 2.99%). No outcome in the test window was visible during fitting.

Metrics are per plate appearance; the probabilities lower down are per game.

Read AUC first — the slate ranks hitters and takes the top six, so ordering is what matters. Brier and log loss are dominated by the ~3% base rate and barely separate the priors.

Prior	AUC	Top-decile lift	Brier	Log loss	Cal. RMSE
Flat league League HR rate as the prior for every hitter.	0.5679	1.48×	0.02901	0.13381	0.00276
KNN comps Prior is the pooled rate of the nearest comparables.	0.5859	1.50×	0.02901	0.13346	0.00459
SVR Support-vector regression on the same feature matrix.	0.5791	1.35×	0.02909	0.13513	0.00886
Random forest Forest regression; a learned similarity metric.	0.5780	1.41×	0.02917	0.13515	0.00914
Core (KNN+SVR) (served) Blend of KNN and SVR, no forest.	0.5859	1.50×	0.02901	0.13346	0.00459
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5859	1.50×	0.02901	0.13346	0.00459
Linear regression Ordinary least squares on the same features.	0.5794	1.24×	0.02918	0.13559	0.01069
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5832	1.41×	0.02917	0.13474	0.00885
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5831	1.30×	0.02911	0.13440	0.00849
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5806	1.45×	0.02913	0.13492	0.00871

Blend weights (forest variant): KNN 1.00, SVR 0.00, RF 0.00

Shrinkage K per prior: league 249, knn 249, svr 249, rf 249, core 249, ensemble 249, linear 249, logistic 249, nn 249, xgb 249 — estimated by method of moments on out-of-fold residuals, not hand-tuned.

Thin-Sample Subset

The same held-out window, restricted to hitters with fewer than 100 plate appearances before the cutoff (4,260 PA, actual HR rate 2.42%). For a regular with 600 PA the observed record swamps any prior, so the priors mostly separate here.

Prior	AUC	Top-decile lift	Brier	Log loss	Cal. RMSE
Flat league League HR rate as the prior for every hitter.	0.5262	1.17×	0.02363	0.11447	0.01072
KNN comps Prior is the pooled rate of the nearest comparables.	0.5387	0.78×	0.02385	0.11701	0.01775
SVR Support-vector regression on the same feature matrix.	0.5551	1.07×	0.02423	0.12431	0.02493
Random forest Forest regression; a learned similarity metric.	0.5456	0.58×	0.02501	0.12635	0.03827
Core (KNN+SVR) (served) Blend of KNN and SVR, no forest.	0.5387	0.78×	0.02385	0.11701	0.01775
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5387	0.78×	0.02385	0.11701	0.01775
Linear regression Ordinary least squares on the same features.	0.5359	0.58×	0.02515	0.13024	0.04004
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5358	0.58×	0.02505	0.12551	0.03866
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5478	0.78×	0.02454	0.12229	0.03088
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5469	1.07×	0.02466	0.12501	0.03169

Slate Agreement

Flat league · 5/6 shared 1/5 hit

Ronald Acuña Jr.
Miguel Vargas
Yordan Alvarez ✓
Pete Alonso
Fernando Tatis Jr.
Cal Raleigh

SVR · 2/6 shared 1/6 hit

Emmanuel Rodriguez
Yordan Alvarez ✓
Yohandy Morales
Miguel Vargas
Matt Olson
Shea Langeliers

Core (KNN+SVR) · 6/6 shared 1/5 hit

Miguel Vargas
Lazaro Montes
Yordan Alvarez ✓
Cal Raleigh
Ronald Acuña Jr.
Pete Alonso

Linear regression · 5/6 shared 1/5 hit

Emmanuel Rodriguez
Lazaro Montes
Miguel Vargas
Yordan Alvarez ✓
Cal Raleigh
Pete Alonso

Neural net (torch) · 4/6 shared 1/6 hit

KNN comps · 6/6 shared 1/5 hit

16.5%	Miguel Vargas	19.3%
15.9%	Lazaro Montes	18.9%
15.8%	Yordan Alvarez ✓	18.6%
15.7%	Cal Raleigh	18.6%
15.5%	Ronald Acuña Jr.	17.8%
15.3%	Pete Alonso	17.8%

Random forest · 4/6 shared 1/6 hit

21.5%	Emmanuel Rodriguez	23.5%
20.3%	Lazaro Montes	21.6%
19.2%	Yordan Alvarez ✓	19.6%
19.2%	Miguel Vargas	18.9%
18.8%	Matt Olson	18.2%
18.7%	Cal Raleigh	18.2%

Forest blend · 6/6 shared 1/5 hit

19.3%	Miguel Vargas	19.3%
18.9%	Lazaro Montes	18.9%
18.6%	Yordan Alvarez ✓	18.6%
18.6%	Cal Raleigh	18.6%
17.8%	Ronald Acuña Jr.	17.8%
17.8%	Pete Alonso	17.8%

Logistic regression · 4/6 shared 1/6 hit

21.5%	Emmanuel Rodriguez	24.2%
21.1%	Yordan Alvarez ✓	20.8%
19.9%	Cal Raleigh	20.2%
19.4%	Miguel Vargas	20.0%
19.0%	Lazaro Montes	19.8%
18.0%	Matt Olson	19.1%

Matt Olson

18.3%

Matt Olson

Matt Olson	18.3%
------------	-------

Hitter by Hitter

[illegible]

Hitter	Flat league	KNN comps	SVR	Random forest	Core (KNN+SVR)	Forest blend	Linear regression	Logistic regression	Neural net (torch)
8. Randal Grichuk Chicago White Sox · vs LHP · 13 HR / 262 PA	14.6%	17.3%	18.5%	16.5%	17.3%	17.3%	16.7%	18.4%	18.0%
9. Matt Olson Atlanta Braves · vs RHP · 40 HR / 675 PA	15.1%	17.2%	18.8%	18.2%	17.2%	17.2%	17.7%	19.1%	18.3%
10. Zach Neto Los Angeles Angels · vs LHP · 26 HR / 675 PA	14.8%	17.0%	16.9%	16.8%	17.0%	17.0%	16.2%	17.1%	16.1%
11. Emmanuel Rodriguez Minnesota Twins · vs RHP · 2 HR / 37 PA	11.3%	16.9%	21.5%	23.5%	16.9%	16.9%	21.5%	24.2%	20.5%
12. Ben Rice New York Yankees · vs LHP · 39 HR / 641 PA	14.7%	16.8%	18.3%	17.7%	16.8%	16.8%	17.3%	17.5%	17.7%
13. Yohandy Morales Washington Nationals · vs RHP · 3 HR / 36 PA	12.1%	16.7%	19.2%	17.2%	16.7%	16.7%	17.5%	17.0%	18.3%
14. Shea Langeliers Athletics · vs LHP · 23 HR / 441 PA	14.0%	16.6%	18.7%	17.7%	16.6%	16.6%	17.2%	18.2%	17.9%
15. Jake Burger Texas Rangers · vs LHP · 27 HR / 596 PA	14.6%	16.6%	17.1%	16.5%	16.6%	16.6%	16.4%	17.4%	17.0%
16. Griffin Conine Miami Marlins · vs RHP · 17 HR / 300 PA	13.8%	16.5%	17.8%	17.2%	16.5%	16.5%	17.3%	17.6%	17.4%
17. Zack Gelof Athletics · vs LHP · 19 HR / 410 PA	14.1%	16.4%	16.7%	16.4%	16.4%	16.4%	17.3%	16.8%	17.0%
18. Tommy Pham Chicago White Sox · vs LHP · 1 HR / 26 PA	14.2%	16.3%	9.9%	16.7%	16.3%	16.3%	16.4%	12.5%	14.6%
19. Fernando Tatis Jr. San Diego Padres · vs RHP · 25 HR / 683 PA	15.5%	16.3%	15.7%	15.2%	16.3%	16.3%	16.4%	16.6%	16.2%
20. Kyle Stowers Miami Marlins · vs RHP · 19 HR / 501 PA	13.3%	16.1%	16.6%	15.9%	16.1%	16.1%	15.2%	16.4%	16.2%
21. Corey Seager Texas Rangers · vs LHP · 18 HR / 412 PA	13.2%	16.0%	15.8%	16.0%	16.0%	16.0%	16.2%	16.2%	16.4%
22. Corbin Carroll	14.2%	16.0%	16.2%	16.0%	16.0%	16.0%	14.9%	15.9%	15.5%

Hitter	Flat league	KNN comps	SVR	Random forest	Core (KNN+SVR)	Forest blend	Linear regression	Logistic regression	Neural net (torch)
37. Luis García Jr. New York Yankees · vs LHP · 30 HR / 527 PA	13.4%	15.0%	16.8%	16.3%	15.0%	15.0%	15.7%	16.4%	16.2%
38. Taylor Trammell Houston Astros · vs RHP · 8 HR / 273 PA	13.1%	15.0%	14.5%	14.9%	15.0%	15.0%	14.7%	15.2%	15.3%
39. Pete Crow-Armstrong Chicago Cubs · vs LHP · 45 HR / 702 PA	13.5%	15.0%	15.9%	15.8%	15.0%	15.0%	15.3%	16.4%	15.8%
40. Ketel Marte Arizona Diamondbacks · vs RHP · 22 HR / 597 PA	14.0%	14.9%	16.6%	14.4%	14.9%	14.9%	15.4%	16.5%	15.8%

Per-game probabilities: each prior's per-PA rate run through this hitter's own plate appearances, park, weather, bullpen and times-through-the-order terms. Only the prior changes across the columns — every other term is held fixed, so a left-to-right spread is the prior disagreeing about the hitter, not the context.